

BIOLOGY ALL IN ONE (BAIO) LEARNING SYSTEM

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BIOLOGICAL CLASSIFICATION

Quick Revision & Concept Summary Notes

1. SYSTEMS OF CLASSIFICATION & HISTORY

- **Aristotle (350 BC):** Earliest scientific attempt.
 - **Plants:** Divided into Trees, Shrubs, and Herbs based on simple morphological characters.
 - **Animals:** Divided into **Enaima** (with red blood) and **Anaima** (without red blood); also classified by habitat into Aquatic and Terrestrial.
- **Two Kingdom System (Linnaeus, 1753):** Plantae and Animalia.
 - **Demerits:** Did not distinguish between *Prokaryotes vs Eukaryotes*, *Unicellular vs Multicellular* (e.g., Chlamydomonas & Spirogyra), or *Photosynthetic vs Non-photosynthetic* (e.g., Green plants with cellulosic cell wall vs Fungi with chitin cell wall). Grouped bacteria, cyanobacteria, fungi, mosses, ferns, gymnosperms & angiosperms together under 'Plants' solely based on having a cell wall.
- **Five Kingdom System (R.H. Whittaker, 1969):** Advanced and most popular system.
 - **Criteria used:** (1) Cell structure, (2) Thallus organization, (3) Mode of nutrition, (4) Reproduction, (5) Phylogenetic relationships.
 - **Merits:** Separated all prokaryotes into Monera; placed all unicellular eukaryotes into Protista (bringing together Chlamydomonas/Chlorella with Paramecium/Amoeba).

2. COMPARATIVE CHARACTERISTICS OF FIVE KINGDOMS

Character	Monera	Protista	Fungi	Plantae	Animalia
Cell Type	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Cell Wall	Noncellular (Polysaccharide + Amino acid)	Present in some	Present (Chitin, without cellulose)	Present (Cellulose)	Absent
Nuclear Membrane	Absent	Present	Present	Present	Present
Body Organization	Cellular	Cellular	Multicellular / Loose tissue	Tissue / Organ	Tissue / Organ / Organ system
Mode of Nutrition	Autotrophic (chemo/photo) & Heterotrophic (saprophytic/ parasitic)	Autotrophic (photo) & Heterotrophic	Heterotrophic (saprophytic / parasitic)	Autotrophic (photosynthetic)	Heterotrophic (Holozoic / Saprophytic)

3. KINGDOM MONERA (PROKARYOTES)

- Sole members are **Bacteria** (most abundant micro-organisms, extreme habitats). Flagella protein = **Flagellin**. Lacks membrane-bound organelles. Cell wall = **Peptidoglycan**.
- **Four Shapes:** Coccus (spherical), Bacillus (rod), Vibrium (comma), Spirillum (spiral).
- **Archaeobacteria (Living Fossils):** Anaerobes surviving extreme conditions due to a unique cell wall structure.
 - *Halophiles*: Salty areas | *Thermoacidophiles*: Hot springs | *Methanogens*: Marshy areas & gut of ruminants (produces Biogas/ Methane).
- **Eubacteria (True Bacteria):** Rigid cell wall + flagellum (if motile).
 - **Photosynthetic Autotrophs:** Cyanobacteria (Blue-green algae). Unicellular/colonial/filamentous with gelatinous sheath. Possess Chlorophyll *a*. Nitrogen fixation in specialized cells called **Heterocysts** (e.g., *Nostoc*, *Anabaena*).
 - **Chemosynthetic Autotrophs:** Oxidize inorganic substances (nitrites, nitrates, ammonia) for ATP generation; recycle N, P, Fe, S.
 - **Heterotrophic Bacteria:** Decomposers. Used in curd production, antibiotics, legume N₂-fixation. Pathogens: Cholera (*Vibrio cholerae*), Typhoid (*Salmonella typhi*), Tetanus, Citrus canker (*Xanthomonas axonopodis*), Anthrax (*Bacillus anthracis*).
- **Mycoplasma:** Smallest living cell, completely lacks cell wall, can survive without oxygen, pathogenic in plants/animals.
- **Reproduction:** Mainly binary fission; spore formation in unfavorable conditions; sexual recombination via DNA transfer (Conjugation).

4. KINGDOM PROTISTA (UNICELLULAR EUKARYOTES)

- Primarily aquatic; flagella/cilia made of protein **tubulin**. Reproduction via cell fusion and zygote formation.
- Chrysophytes**: Diatoms & Golden algae (Desmids). Freshwater/marine, photosynthetic. Cell walls form 2 thin overlapping shells embedded with silica (indestructible). Forms **Diatomaceous Earth** (used in polishing, filtration of oils/syrups, soundproofing). Chief producers in oceans.
- Dinoflagellates**: Mostly marine & photosynthetic. Cell wall has stiff cellulosic plates. 2 flagella (one longitudinal, one transverse in a wall furrow). Red dinoflagellates (*Gonyaulax*) multiply rapidly causing **Red Tides** and release toxins killing marine life.
- Euglenoids**: Freshwater stagnant water organisms. Lack cell wall, contain protein-rich layer called **Pellicle** (flexible). 2 flagella (short & long). Mixotrophic nutrition (Autotrophic in light, Heterotrophic/predatory in dark). Connecting link between plants and animals.
- Slime Moulds**: Saprophytic protists. Form aggregation called **Plasmodium** in suitable conditions. Under unfavorable conditions, plasmodium forms fruiting bodies with resistant spores dispersed by air currents.
- Protozoans**: Heterotrophs (predators/parasites).
 - Amoeboid*: Pseudopodia for movement/capture (e.g., *Amoeba*, *Entamoeba* - parasite). Marine forms have silica shells.
 - Flagellated*: Free-living or parasitic. Cause sleeping sickness (*Trypanosoma*), Leishmaniasis/Black fever (*Leishmania*), *Giardia*.
 - Ciliated*: Actively moving with thousands of cilia & gullet (e.g., *Paramecium*).
 - Sporozoans*: Infectious spore-like stage (e.g., *Plasmodium* - Malaria parasite).

5. KINGDOM FUNGI

- Achlorophyllous, heterotrophic (saprophytic/parasitic). Multicellular & filamentous (except unicellular **Yeast**). Thread-like structures called **Hyphae**; network of hyphae = **Mycelium**. Multinucleate continuous hyphae = **Coenocytic**. Cell wall = **Chitin** + polysaccharides.
- Symbiosis**: Lichens (with algae) & Mycorrhiza (with roots of higher plants).
- Reproduction**: Vegetative (fragmentation, fission, budding); Asexual (Conidia, Sporangiospores, Zoospores); Sexual (Oospores, Ascospores, Basidiospores).
 - Sexual Steps*: (1) Plasmogamy (fusion of protoplasm) → (2) Karyogamy (fusion of nuclei) → (3) Meiosis in zygote.
 - Dikaryon Stage (n + n)**: Intervening phase before karyogamy in Ascomycetes and Basidiomycetes.

Class	Mycelium	Asexual Spores	Sexual Spores & Fruiting Body	Key Examples & Notes
Phycomycetes (Algal Fungi)	Aseptate & Coenocytic	Zoospores (motile) / Aplanospores (non-motile) [Endogenous]	Zygospores (Isogamous / Anisogamous / Oogamous)	<i>Mucor</i> , <i>Rhizopus</i> (Bread mould), <i>Albugo</i> (parasite on mustard). Aquatic/damp habitats.
Ascomycetes (Sac Fungi)	Branched & Septate	Conidia [Exogenous on Conidiophores]	Ascospores [Endogenous in Asci & Ascocarps]	<i>Yeast</i> (unicellular), <i>Aspergillus</i> , <i>Claviceps</i> , <i>Neurospora</i> (genetics), <i>Penicillium</i> (antibiotic). Morels & Truffles (edible). Coprophilous (on dung).
Basidiomycetes (Club Fungi)	Branched & Septate	Absent (Vegetative via Fragmentation)	Basidiospores [Exogenous on Basidia & Basidiocarps]	<i>Agaricus</i> (Mushroom), <i>Ustilago</i> (Smut), <i>Puccinia</i> (Rust), Bracket fungi. Sex organs absent; Plasmogamy via somatic cells.
Deuteromycetes (Imperfect Fungi)	Branched & Septate	Conidia	Absent / Unknown (Sexual stage missing)	<i>Alternaria</i> (Early blight of potato), <i>Colletotrichum</i> (Red rot of sugarcane), <i>Trichoderma</i> . Decomposers aiding mineral cycling.

6. KINGDOM PLANTAE & ANIMALIA OVERVIEW

- Plantae**: Eukaryotic, photosynthetic, cellulosic cell wall, chloroplasts. Alternation of generation (Sporophytic 2n & Gametophytic n). Includes Algae, Bryophytes, Pteridophytes, Gymnosperms, Angiosperms.
 - Insectivorous / Partial Heterotrophs*: Venus flytrap (*Dionaea*), Bladderwort, Pitcher plant (*Nepenthes*), *Sarracenia*, Sundew (*Drosera*). Parasite: *Cuscuta*.
- Animalia**: Heterotrophic, eukaryotic, multicellular, no cell wall, internal digestion, glycogen/fat storage, direct/indirect plant dependency, capable of locomotion.

7. SUB-VIRAL AGENTS & SYMBIONTS

- Viruses**: Acellular, inert crystalline structure outside living host. Obligate parasites. Genetic material: Either DNA or RNA (never both), surrounded by a protein coat called **Capsid** made of subunits called **Capsomeres**.
 - Plant Viruses*: Single-stranded RNA (e.g., TMV).
 - Bacteriophages*: Double-stranded DNA.

- *Animal Viruses*: ssRNA, dsRNA, or dsDNA.
- *Discovery Timeline*: Term by Pasteur → D.J. Ivanowsky (1892, mosaic disease, filter-passing) → M.W. Beijerinck (1898, *Contagium vivum fluidum*) → W.M. Stanley (1935, crystallized virus).
- *Virion*: Completely assembled infectious virus particle outside host.
- **Viroids**: Discovered by **T.O. Diener (1971)**. Smaller than viruses, **lacks protein coat** (free low molecular weight RNA, $\sim 10^5$ Daltons). Causes *Potato Spindle Tuber Disease (PSTVd)* and *Chrysanthemum Stunt Disease*.
- **Prions**: Infectious abnormally folded proteins of size similar to viruses. Cause neurological diseases: *Bovine Spongiform Encephalopathy (BSE / Mad Cow)* in cattle and *Creutzfeldt-Jakob Disease (CJD)* in humans.
- **Lichens**: Symbiotic association between Algae (**Phycobiont** - autotrophic) and Fungi (**Mycobiont** - heterotrophic). Algae provides food; fungi provides shelter, water, and minerals. Excellent pollution indicators (do not grow in SO₂ polluted areas).